(Please write your Enrolment No. immediately)

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

6th Semester, April, 2024

Paper Code: DA-304T **Subject:** Statistics, Statistical Modelling and Data Analytics(SSMD)

Time:1 ½ Hrs. **Max.Marks: 30**

Note: Attempt Q.No. 1 which is compulsory and any two more questions from remaining.

Q.No.		Question	Max.Marks	CO(s)
1	a)	Three students A, B, and C are running in a race. A and B have the same probability of winning and each is twice as likely to win as C. Find the probability that B or C wins	3M	CO1
	b)	Define Properties of the Normal Distribution and also Define any two hypothesis tests	2M	CO1
	c)	Obtain the line of regression of y on x for the data given below:	3.5M	CO2

X	1.53	1.78	2.60	2.95	3.42
Y	33.50	36.30	40.00	45.80	53.50

	d)	Define Data Visualization and Least square method with a real life application						1.5M	CO2
2	a)	Find the mean and variance of uniform probability distribution $f(x)=\frac{1}{n}$ for x=1,2,...n						5M	CO1
	b)	For discrete probability distribution						5M	CO1

x	0	1	2	3	4	5	6	7
f	0	k	2k	2k	3k	K^2	$2k^2$	$7k^2+k$

Determine: (i)k (ii) Mean, (iii) Variance (iv) smallest value of x such that $P(X \le x) > \frac{1}{2}$

3	a)	State and prove Gauss Markov theorem	8M	CO2
	b)	Let V be a vector space in $\mathbb{R}^3$. Examine whether the following are the subspaces or not? (i) $W=\{(a,b,c) \mid a^2+b^2+c^2 \le 1\}$ and (ii) $W=\{(a,b,c) \mid a=b=c\}$	2M	CO2
4	a)	Fit a straight line to the following data considering y as the dependent variable:	8M	CO2

x	1	2	3	4	5
y	5	7	9	10	11

	b)	In a sample of 1000 people, 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice eater and wheat eater are equally popular at 1% level of significance	2M	CO1

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